

Where Diffusion Counterfactuals Stop Explaining and Start Repainting: An Operating-Point Analysis on Abdominal CT

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Abstract—Counterfactual explanation asks what would have to change for a classifier to decide otherwise, and diffusion models are an increasingly common way to generate the answer. The method has a free parameter that is rarely studied: how far up the forward diffusion trajectory the original image is taken before guided denoising begins. We train an unconditional denoising diffusion probabilistic model on 34,561 abdominal CT slices, attach a classifier-guided sampling procedure, and sweep this noising depth across seven values while measuring validity, proximity, pixel-level sparsity and identity preservation on 256 held-out slices. Validity does not increase monotonically with noising depth. It peaks at 0.426 and then declines, while identity correlation with the original slice collapses from 0.981 to 0.079. Past the peak the procedure changes 88% of pixels and retains almost nothing of the input, so whatever it produces is no longer an explanation of that slice. We additionally run a nearest-neighbour memorisation audit against the training set and find no evidence of training-image reproduction, with generated samples no closer to training data than real held-out images are. We argue that reporting a single noising depth without the surrounding curve makes diffusion counterfactuals difficult to interpret, and that the memorisation audit should be routine for generative models trained on patient data. Code and all figures are public.

Index Terms—explainable AI, counterfactual explanation, diffusion models, computed tomography, generative models, memorisation

I. INTRODUCTION

Saliency maps tell a reader where a network looked. They do not say what would have had to be different for the network to reach another conclusion. A counterfactual explanation answers the second question directly: given an image the model labels y , produce the nearest image the model labels y' . If the edit lands on structures a clinician would also identify, the explanation is informative; if it scatters changes across the whole field of view, it is not.

Diffusion models are well suited to generating such edits because they define a continuous path between data and noise. The standard recipe noises a real image forward to some timestep t_{start} and denoises it back down under gradient guidance from the classifier being explained [1], [2]. The choice of t_{start} controls a trade-off that is intuitively obvious and, in practice, rarely reported: too little noise leaves the image unchanged and the prediction unmoved, too much destroys the evidence the explanation was supposed to be about.

This paper treats that parameter as the object of study rather than a hyperparameter to be fixed. Our contributions are:

- 1) A systematic sweep of noising depth on abdominal CT, measuring validity, proximity, sparsity and identity on the same 256 slices, showing that validity is *non-monotonic* and peaks well before the image is destroyed.
- 2) Identification of an operating point beyond which the procedure stops being an explanation of the input, with quantitative criteria for locating it.
- 3) A nearest-neighbour memorisation audit with a real-data control, which we argue should accompany any medical generative model, following the finding that latent diffusion models can reproduce training patients [5].

II. RELATED WORK

Counterfactual explanation. DiME [1] introduced diffusion-based counterfactual generation for natural images. In medical imaging, diffusion autoencoders have been used for classification and regression explanations [2], and counterfactual diffusion has been applied to histopathology. Salahuddin et al. [3] combine counterfactual explanation with uncertainty estimation for renal cyst segmentation on contrast-enhanced abdominal CT using a VAE-GAN, and validate across centres. The present work substitutes a diffusion model for the VAE-GAN and studies the sampling parameter that substitution introduces.

Interpretability evaluation. Salahuddin et al. [4] survey interpretability for medical image analysis and note that explanation quality is frequently asserted rather than measured. We follow that recommendation by reporting four quantitative axes rather than a gallery of examples.

Memorisation. Dar et al. [5] show that unconditional latent diffusion models trained on medical images can reproduce training patients, with direct implications for sharing synthetic data. We adopt their framing and report a nearest-neighbour statistic against a real-data control.

III. METHOD

A. Data

We use OrganAMNIST from MedMNIST v2 [6], the 64×64 release. The dataset contains axial abdominal CT

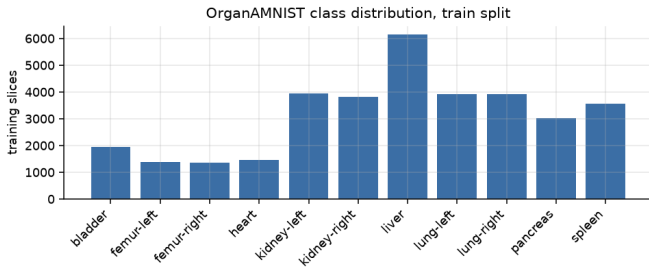


Fig. 1. OrganAMNIST training class distribution. The imbalance motivates per-class reporting in Table I.

slices cropped to eleven organ classes and derived from the Liver Tumor Segmentation Benchmark. Splits are 34,561 training, 6,491 validation and 17,778 test images. Intensities are mapped from uint8 to $[-1, 1]$. Fig. 1 shows the class distribution, which is uneven: liver contributes 6,164 training slices and femur-left 1,390. We therefore report per-class classifier accuracy rather than a single average.

B. Generative model

The diffusion model is a MONAI DiffusionModelUNet with 10.4M parameters, channel widths (64, 128, 192), two residual blocks per level and self-attention at the coarsest resolution only. It is trained to predict the noise ϵ added at timestep t under a DDPM schedule with $T = 1000$ [7], minimising

$$\mathcal{L} = \mathbb{E}_{x_0, \epsilon, t} \|\epsilon - \epsilon_\theta(\sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon, t)\|_2^2. \quad (1)$$

Training ran for 40 epochs at batch size 512 in bfloat16, taking 23 minutes on one NVIDIA A100.

C. Classifier under explanation

The model being explained is a four-stage convolutional network over 64×64 slices with 11 outputs. It is deliberately small: the object of study is the explanation procedure, and a larger classifier would slow the counterfactual search without changing what is measured.

D. Counterfactual generation

Given a slice x_0 with source label y and a target label $y' \neq y$, we noise x_0 forward to t_{start} and denoise with DDIM [8] over 100 steps. At each step we form the predicted clean image

$$\hat{x}_0^{(t)} = \frac{x_t - \sqrt{1 - \alpha_t} \epsilon_\theta(x_t, t)}{\sqrt{\alpha_t}}, \quad (2)$$

and modify the noise prediction using the classifier gradient,

$$\tilde{\epsilon} = \epsilon_\theta(x_t, t) - s\sqrt{1 - \alpha_t} \nabla_{\hat{x}_0} \log p_\phi(y' | \hat{x}_0^{(t)}), \quad (3)$$

with guidance scale $s = 6$.

Taking the gradient with respect to $\hat{x}_0^{(t)}$ rather than x_t is deliberate. The classifier was trained on clean slices, so evaluating it on a partially noised latent queries it about inputs outside its training distribution.

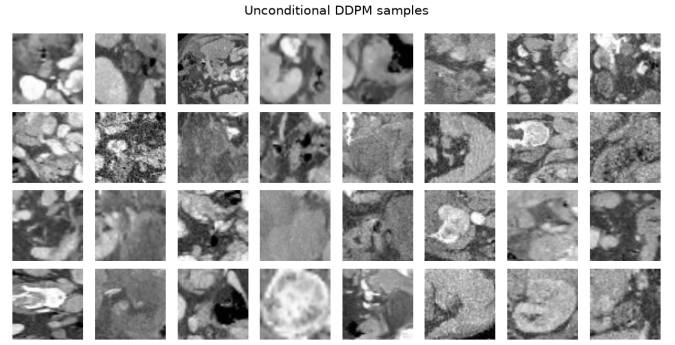


Fig. 2. Unconditional samples after 40 epochs. Organ boundaries, vessels, bright contrast-enhanced structures and dark air appear in plausible arrangements.

TABLE I
CLASSIFIER TEST ACCURACY PER ORGAN

Organ	Acc.	Organ	Acc.
heart	0.885	pancreas	0.976
kidney-left	0.899	spleen	0.982
femur-left	0.946	liver	0.992
kidney-right	0.946	lung-right	0.999
bladder	0.952	lung-left	1.000
femur-right	0.968	overall	0.958

E. Evaluation

We report four axes on 256 randomly chosen test slices, on which the classifier achieves 0.934 accuracy. **Validity** is the fraction for which the classifier predicts y' after editing. **Proximity** is mean L_1 change and the fraction of pixels moving more than 0.1 on the $[-1, 1]$ scale. **Identity** is the centred cosine similarity between x_0 and the counterfactual. **Realism** is a Frechet distance computed on features of the organ classifier rather than ImageNet Inception, because Inception features are trained on natural photographs and describe CT poorly; the number is therefore internally comparable only.

For memorisation we compute, for each of 1,024 generated samples, the minimum L_2 pixel distance to any of 2,048 training images, and compare the distribution against the same statistic computed for real held-out test images.

IV. RESULTS

A. Sample quality

Fig. 2 shows unconditional samples. Noise-prediction MSE fell from 0.096 to 0.034 over training. The Frechet distance between generated samples and training data is 46.8 in classifier feature space, against a floor of 2.9 measured between real test images and training data. Samples remain distinguishable from real data, which is expected at this training budget.

B. Classifier

The classifier reaches 95.8% test accuracy. Per-class results (Table I) span 0.885 for heart to 1.000 for left lung; lungs are near-perfect because they present air against soft tissue.

TABLE II
COUNTERFACTUAL QUALITY AGAINST NOISING DEPTH, 256 TEST SLICES

t_{start}	Validity	L_1	Pixels chg.	Identity
50	0.086	0.054	0.162	0.981
100	0.301	0.090	0.344	0.950
150	0.387	0.114	0.424	0.923
200	0.426	0.155	0.546	0.866
300	0.375	0.305	0.777	0.602
400	0.398	0.422	0.851	0.335
600	0.219	0.481	0.876	0.079

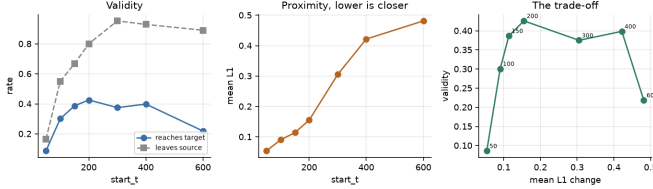


Fig. 3. Validity, proximity, and the trade-off between them. Validity peaks at $t_{\text{start}} = 200$ and declines thereafter while the L_1 change continues to grow.

C. Validity is non-monotonic in noising depth

Table II and Fig. 3 give the main result.

Validity rises steeply to 0.426 at $t_{\text{start}} = 200$, then falls to 0.219 at $t_{\text{start}} = 600$. Meanwhile L_1 change grows monotonically from 0.054 to 0.481 and identity correlation decays from 0.981 to 0.079.

The common assumption that deeper noising buys more successful flips is therefore false in this setting. Past the peak, guided denoising has enough freedom to leave the neighbourhood of the original slice entirely. The classifier is then being asked about a different image, not about a modified version of the input, and it is no longer obliged to move toward the target class. At $t_{\text{start}} = 600$ the procedure changes 88% of pixels and retains essentially nothing of the input.

We suggest a simple operating criterion: choose the largest t_{start} for which identity correlation remains above a threshold (here 0.85), and verify that validity has not yet begun to fall. Both conditions select $t_{\text{start}} = 200$.

Fig. 4 shows qualitative results at the selected operating point.

D. Memorisation audit

Generated samples sit approximately 10% closer to the training set than real unseen images (Table III), and the closest generated sample is *further* from training data than the closest real test image. There is no evidence of training-image reproduction at this scale. We stress that this audit costs minutes and should accompany any medical generative model whose outputs may be shared.

V. LIMITATIONS

Results are on 64×64 single slices rather than full CT volumes, a deliberate trade to keep the pipeline within a single-GPU day. The method transfers but the numbers should not be quoted as volumetric results. Peak validity is 0.426, so

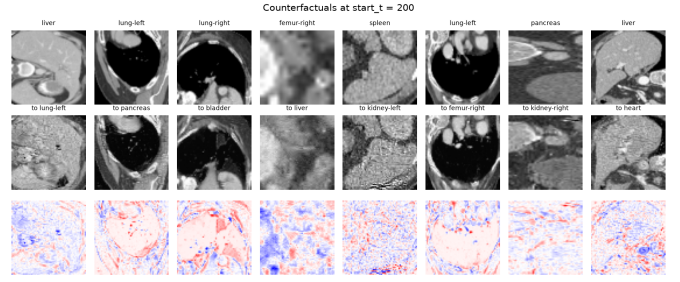


Fig. 4. Counterfactuals at the operating point. Top: original. Middle: counterfactual. Bottom: difference. Anatomy is preserved and edits are localised and structured rather than uniform noise.

TABLE III
NEAREST-NEIGHBOUR DISTANCE TO THE TRAINING SET

Query set	Median	Minimum	Ratio
Generated ($n=1024$)	22.68	11.48	0.899
Real test ($n=310$)	25.22	10.71	1.000

most counterfactuals fail; part of this is the genuine difficulty of converting one organ’s field of view into another’s, which no guidance should achieve, and a conditional diffusion model would likely improve the remainder. Frechet distance is measured in classifier feature space and is not comparable to published FID values. Organ identity is a proxy for a clinical label; applying the pipeline to a diagnostic target such as malignancy is the natural next step. Finally, the sweep was run with a single seed, so the shape of the curve is reliable but individual values carry sampling noise.

VI. CONCLUSION

Noising depth is not a hyperparameter to be fixed and forgotten in diffusion counterfactual explanation. Validity is non-monotonic in it, and beyond a locatable operating point the procedure stops explaining the input and begins replacing it. Reporting the curve rather than a single setting makes the resulting explanations interpretable. We further find no evidence of patient memorisation in our generative model, using an audit we recommend as routine.

REPRODUCIBILITY

Code, trained-model configuration, result files and all figures are available at <https://github.com/Joana-Mansa/ct-diffusion-counterfactuals>. The full pipeline runs in approximately 35 minutes on one A100 and downloads its data automatically.

REFERENCES

- [1] G. Jeanneret, L. Simon, and F. Jurie, “Diffusion models for counterfactual explanations,” *arXiv:2203.15636*, 2022.
- [2] P. Sanchez et al., “Counterfactual explanations for medical image classification and regression using diffusion autoencoder,” *J. Machine Learning for Biomedical Imaging*, 2024.
- [3] Z. Salahuddin et al., “Counterfactuals and uncertainty-based explainable paradigm for the automated detection and segmentation of renal cysts in computed tomography images: a multi-center study,” *arXiv:2408.03789*, 2024.

- [4] Z. Salahuddin, H. C. Woodruff, A. Chatterjee, and P. Lambin, "Transparency of deep neural networks for medical image analysis: a review of interpretability methods," *Computers in Biology and Medicine*, vol. 140, 2022.
- [5] S. U. Dar et al., "Unconditional latent diffusion models memorize patient imaging data: implications for openly sharing synthetic data," *arXiv:2402.01054*, 2024.
- [6] J. Yang et al., "MedMNIST v2: a large-scale lightweight benchmark for 2D and 3D biomedical image classification," *Scientific Data*, vol. 10, 2023.
- [7] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *NeurIPS*, 2020.
- [8] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *ICLR*, 2021.